

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192424055
Date	08 October 2026
Day	Thursday

SET 13: Geographic Data Analysis

Aim

To analyze geographic data using a **map chart, scatter plot, table, and Tableau dashboard** for exploring the relationship between population, average temperature, and elevation.

Dataset

Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500,000	75	1,000
City B	700,000	68	800
City C	600,000	80	1,200

Q1. Map Chart — Distribution of Cities

Question

Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

R Program

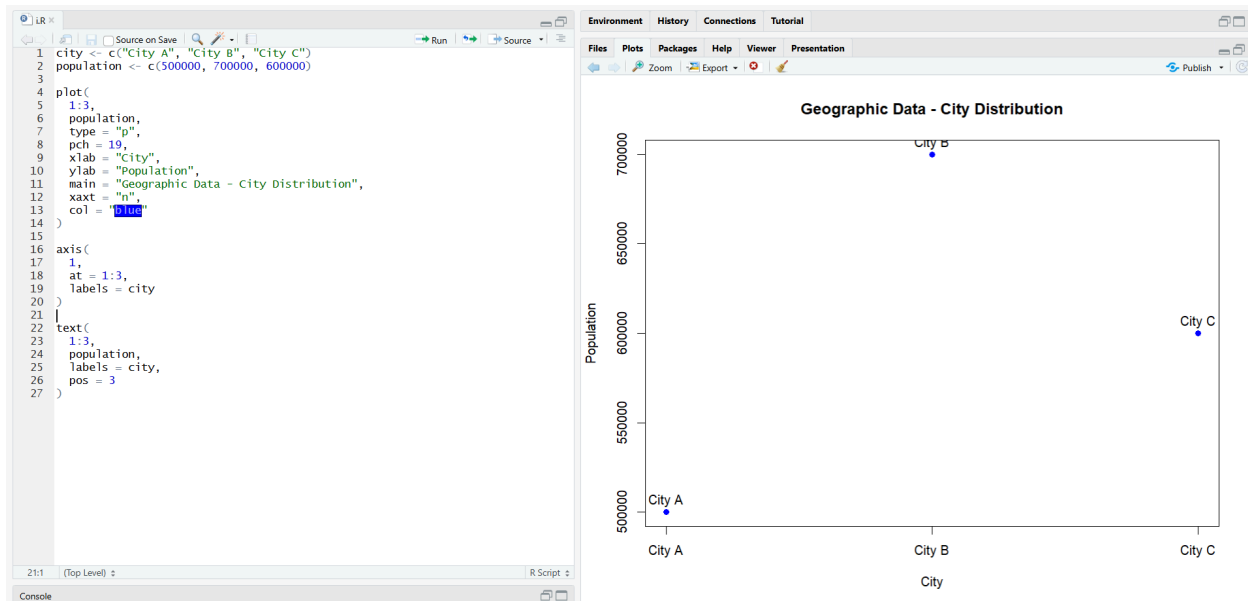
```
city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)

plot(
  1:3,
  population,
  type = "p",
  pch = 19,
  xlab = "City",
  ylab = "Population",
  main = "Geographic Data - City Distribution",
  xaxt = "n",
  col = "blue"
)

axis(
  1,
  at = 1:3,
  labels = city
)

text(
  1:3,
  population,
  labels = city,
  pos = 3
)
```

Output



Observation

The dataset contains three cities with different population sizes. **City B has the largest population**, while **City A has the smallest**.

Q2. Scatter Plot — Temperature vs Population

Question

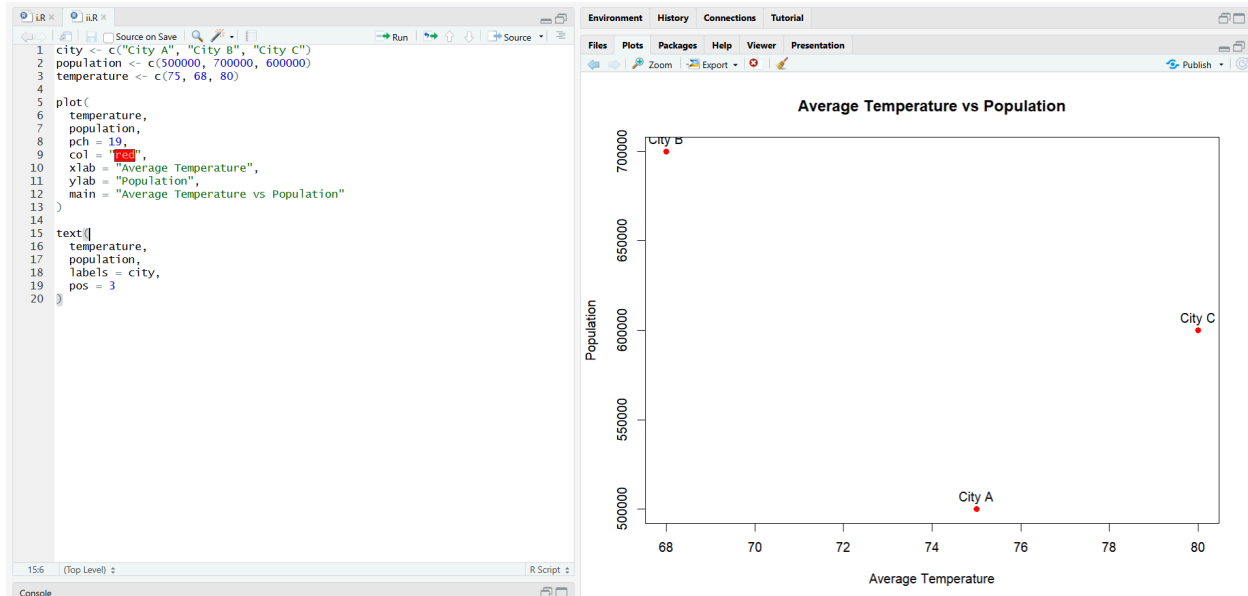
Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

R Program

```
city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
temperature <- c(75, 68, 80)
plot(
  temperature,
  population,
  pch = 19,
  col = "red",
  xlab = "Average Temperature",
  ylab = "Population",
  main = "Average Temperature vs Population"
)

text(
  temperature,
  population,
  labels = city,
  pos = 3
)
```

Output



Observation

- City A: Temperature = **75**, Population = **300,000**
- City B: Temperature = **68**, Population = **700,000**
- City C: Temperature = **80**, Population = **600,000**

The three observations do not show a clear linear relationship between temperature and population. With only three cities, no strong conclusion about their general relationship can be made.

Q3. Geographic Data Table

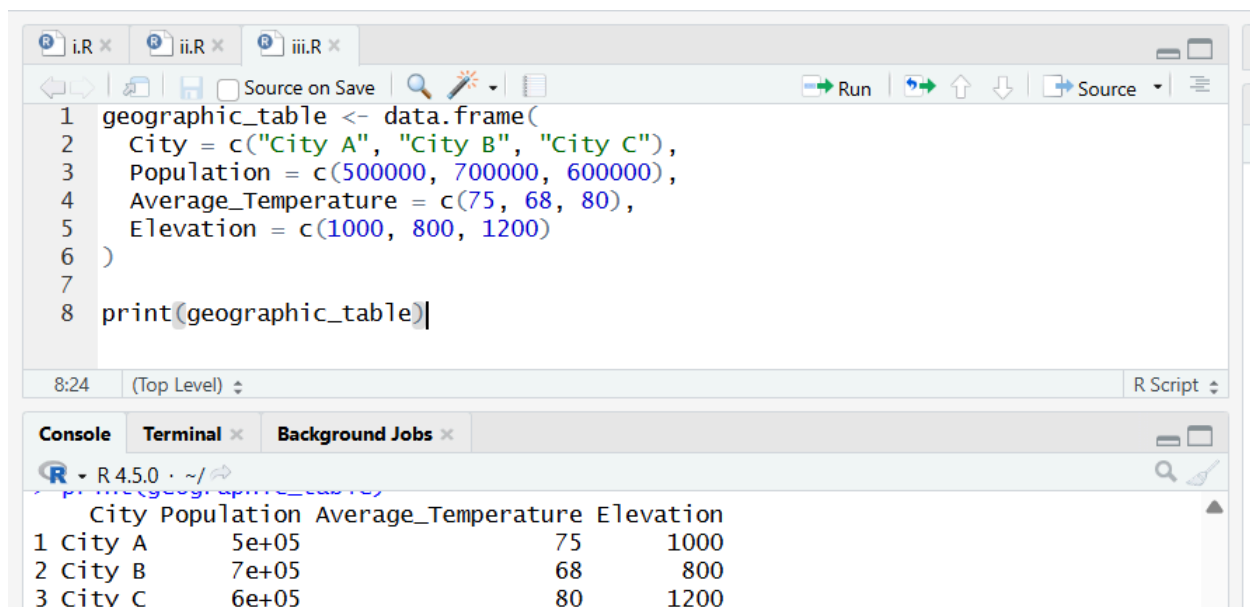
Question

Build a table to display the geographic data for each city. Label the table elements.

R Program

```
geographic_table <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Population = c(500000, 700000, 600000),  
  Average_Temperature = c(75, 68, 80),  
  Elevation = c(1000, 800, 1200)  
)  
  
print(geographic_table)
```

Output



The screenshot shows the R Studio environment. The top pane displays the R script with the following code:

```
1 geographic_table <- data.frame(  
2   City = c("City A", "City B", "City C"),  
3   Population = c(500000, 700000, 600000),  
4   Average_Temperature = c(75, 68, 80),  
5   Elevation = c(1000, 800, 1200)  
6 )  
7  
8 print(geographic_table)|
```

The bottom pane shows the console output of the script:

```
print(geographic_table)  
  City Population Average_Temperature Elevation  
1 City A      5e+05             75         1000  
2 City B      7e+05             68           800  
3 City C      6e+05             80         1200
```

Observation

The table successfully displays the complete geographic information for all three cities.

Q4. Tableau Dashboard

Question

Develop a Tableau dashboard combining the **map chart, scatter plot, and table** for interactive exploration of geographic data.

Step 1 — Prepare the Dataset

Create an Excel or CSV file containing:

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Step 2 — Import into Tableau

1. Open **Tableau**.
2. Select **Microsoft Excel** or **Text File**.
3. Import the geographic dataset.

Step 3 — Create Map Chart

If the cities are recognized as real geographic locations:

1. Double-click **City**.
2. Tableau will attempt to generate a map.
3. Drag **Population** → Size.
4. Drag **City** → Label.
5. Add **Elevation** or **Average Temperature** to Color if required.

Title: Geographic Distribution of Cities

With the labels **City A**, **City B**, **City C**, Tableau may not recognize them as actual geographic locations. In that case, latitude and longitude must be supplied to create an accurate geographic map.

Step 4 — Create Scatter Plot

1. Create a new worksheet.
2. Drag **Avg. Temperature** → Columns.
3. Drag **Population** → Rows.
4. Drag **City** → Label.
5. Select **Circle** from the Marks card.

Title: Average Temperature vs Population

Step 5 — Create Data Table

Add:

- City
- Population
- Avg. Temperature
- Elevation

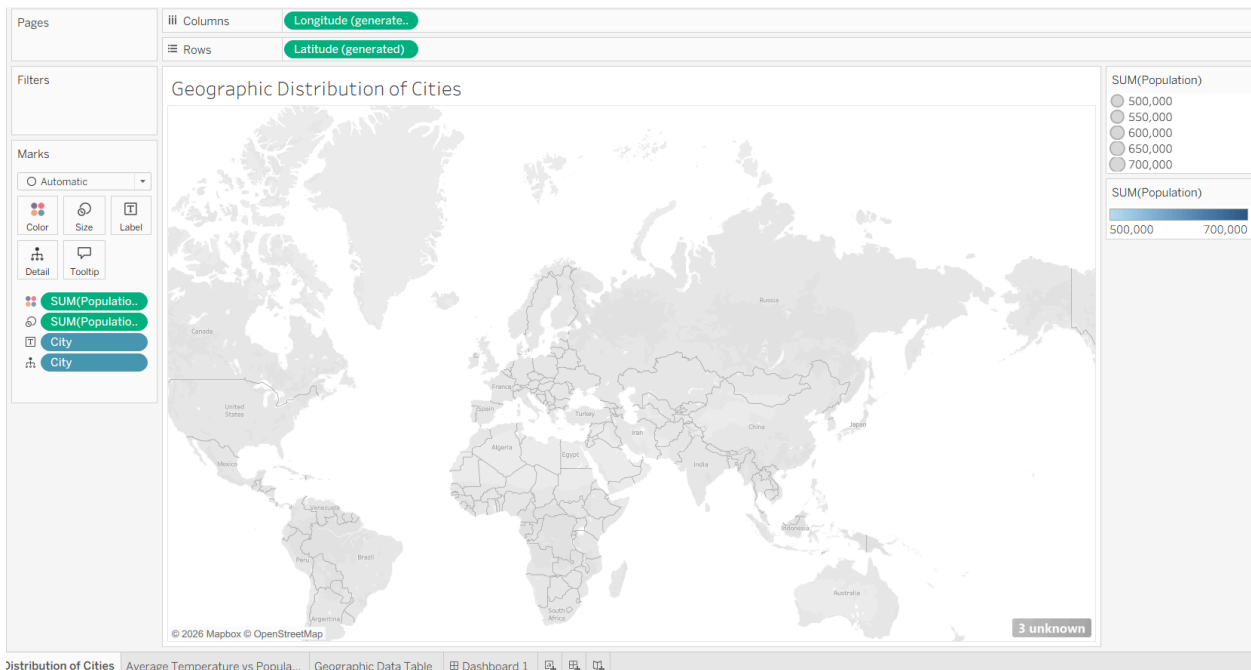
to the worksheet.

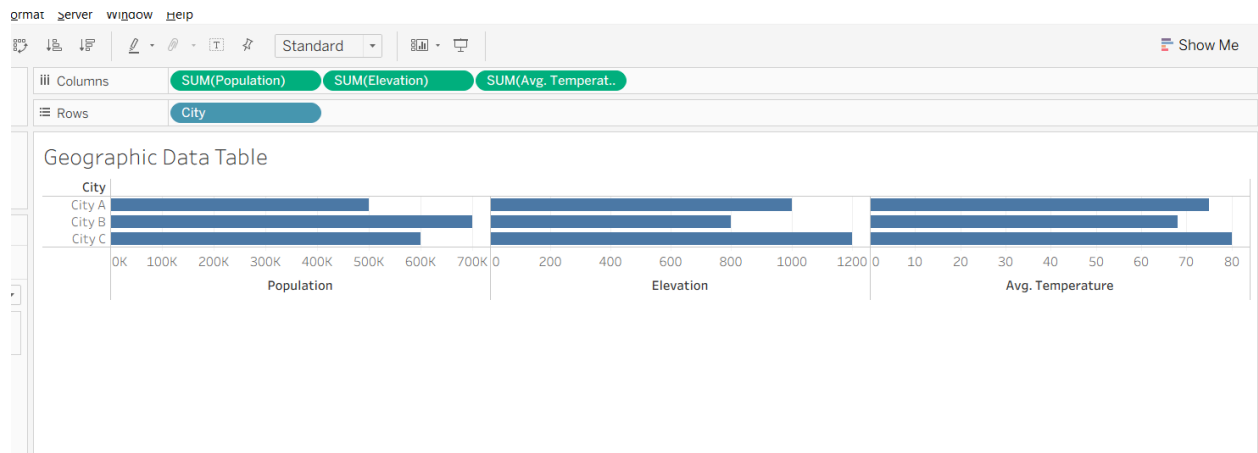
Title: Geographic Data Table

Step 6 — Create Dashboard

1. Select **New Dashboard**.
2. Add the map chart.
3. Add the scatter plot.
4. Add the geographic data table.
5. Add filters such as City if required.
6. Arrange the visualizations neatly.

Expected Dashboard

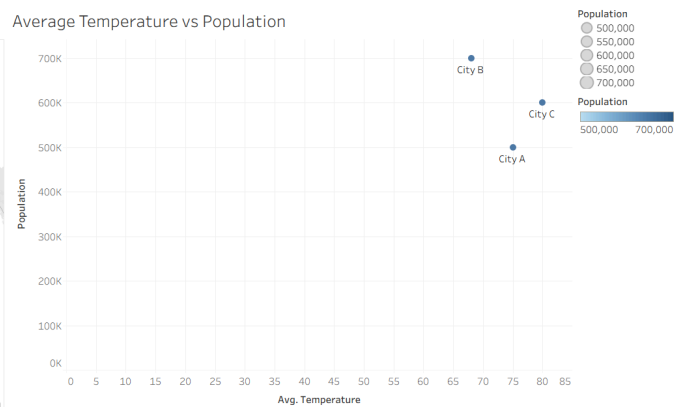




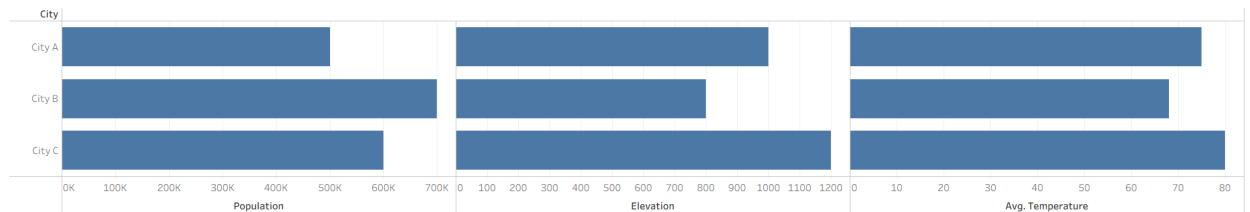
Geographic Distribution of Cities



Average Temperature vs Population



Geographic Data Table



Procedure

1. Open **RStudio** and enter the given geographic dataset.
2. Create a city distribution visualization using the available city and population information.
3. Create a scatter plot using **Average Temperature** and **Population**.
4. Add city names as labels to the scatter plot.
5. Use `data.frame()` to create a table containing City, Population, Average Temperature, and Elevation.
6. Open **Tableau** and import the geographic dataset.
7. Create a map chart using geographic information.
8. Create a scatter plot of Average Temperature versus Population.
9. Create a table displaying the geographic data for each city.
10. Create a new Tableau dashboard.
11. Combine the **map chart, scatter plot, and table** into the dashboard.
12. Add suitable labels, titles, filters, and formatting for interactive exploration.

Result

The geographic dataset was successfully analyzed using **R and Tableau**. A city distribution visualization, temperature-versus-population scatter plot, and geographic data table were created. These visualizations were combined into a **Tableau dashboard** for interactive exploration of the geographic data.